MEASURED RISC-V SOC PORTFOLIO

From a five-stage RV32I + Zicsr core to a verified two-wide out-of-order SoC with an int8 neural accelerator

2-WIDE OOO
5-STAGE
M9K MEMORY
4x4 NPU

LOCKSTEP / ASSERT / MEASURE

+20.91%

CoreMark/MHz gain

1.176568 in-order to 1.422552 OoO on the same 720-iteration image

MEASURED SIMULATION

99.3%

RTL line coverage

1596 of 1607 lines across 61 programs per core with zero failures

VERIFICATION EVIDENCE

93.30x

OoO NPU cycle speedup

58,535,712 software cycles to 627,343 NPU cycles; logits bit-exact

MEASURED SIMULATION

25 MHz

Timing-clean board build

31.29 MHz Fmax and +8.045 ns slow-85C setup slack; not yet flashed

QUARTUS BUILD

Hanna Ashkar

Hanna Ashkar - Electrical Engineering, Technion

https://github.com/hannaashkar/rv32i_cpu

BUILD-PROVEN 25 MHz / SILICON-PROVEN 16.67 MHz

SYSTEM ARCHITECTURE

One software stack, two CPUs, one measured SoC

The same freestanding C and assembly binaries run on the baseline and OoO cores. Both retire into an independent project-ISA model, while shared M9K memories, MMIO, and the NPU keep the platform comparison honest.

Bare-metal RV32I + Zicsr software

IDENTICAL BINARY

Five-stage in-order CPU

IF / ID / EX / MEM / WB

- Forwarding, interlocks, branch prediction
- Cycle and instret hardware counters
- 50 MHz hardware-confirmed

Two-wide out-of-order CPU

R10K-style rename and speculative execution

- 64 physical registers / 32-entry ROB
- 16-entry IQ / 8-entry SQ / 8-entry LQ
- Gshare + BTB + RAS + store sets
- Two-wide dispatch and retirement

Shared SoC platform

M9K memory system

Banked instruction memory and initialized data memory

MMIO platform

LEDs, switches, six seven-segment displays, and test/console hooks

4x4 int8 NPU

Output-stationary systolic array with 16 hardware multipliers

Independent ISA model

Every retired PC, register writeback, and memory store compared

RTL retirement

same binary / same ISA contract

Terasic DE10-Lite / Intel MAX 10 / PLL + SDC / internal-flash boot

PERFORMANCE

Performance journey: IPC gains meet FPGA physics

The OoO core wins normalized CPU work, but its first board port exposed the central physical-design lesson: wall-clock performance is IPC multiplied by clock frequency.

Current CoreMark/MHz A/B

Five-stage in-order

1.176568

Two-wide OoO D029

1.422552

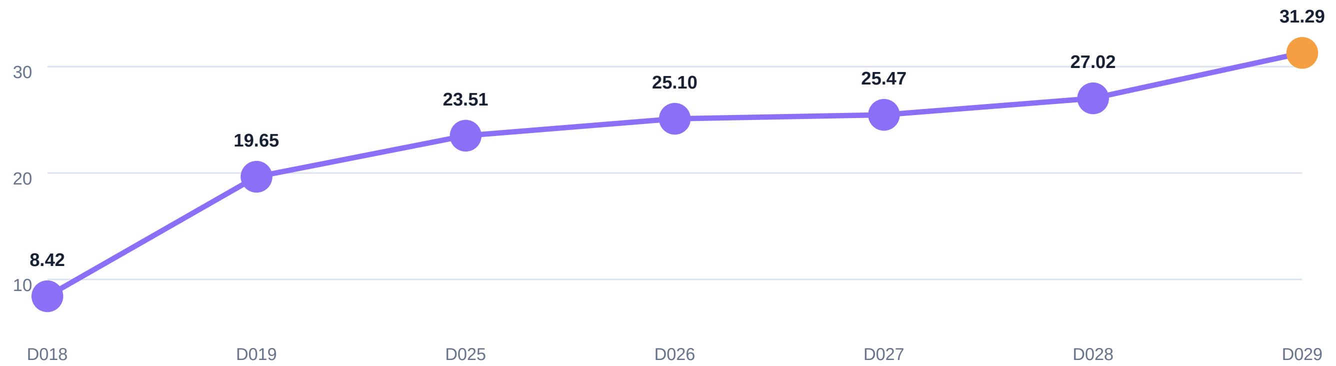
+20.91% ON THE EXACT SAME 720-ITERATION IMAGE

Reportable D029 CoreMark

IPC	1.026
Full-run cycles	506,197,207
Retired + checked	519,453,600
Iterations	720

Official CRCs pass; Correct operation validated; zero lockstep divergence

Routed OoO Fmax journey



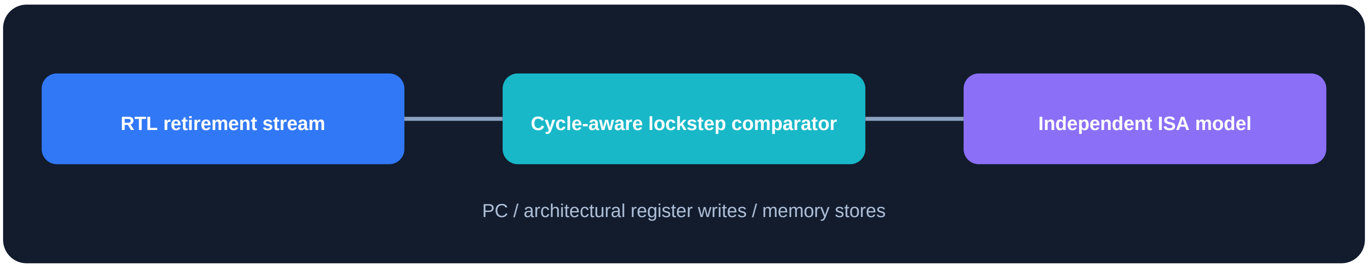
The honest result

The hardware-confirmed in-order core runs at 50 MHz, while the hardware-confirmed OoO image runs at 16.67 MHz. D029 closes a 25 MHz image with margin, but it remains unflashed. The project reports the IPC win and the remaining frequency gap together.

VERIFICATION

Verification is part of the microarchitecture

Every CPU test and benchmark runs retirement lockstep by default. Directed tests explain known corners; independent models, random stress, assertions, and coverage are the safety nets for unknown ones.



1596 / 1607 RTL lines

61 programs per core

21/21
Directed assembly + C
Both cores

40/40
riscv-tests rv32ui
Both cores

25/25
Base / system / NPU random
Each lane, each core

25/25
Load-violation stress
OoO only

84/84
Randomized X/reset runs
Each core

15 documented RTL / SoC bugs

B009

Slot-1-only store silently missed SQ allocation

Lockstep within minutes of first OoO boot

B011

Stalled NPU access recomputed operands from decaying bypasses

Adversarial review before merge

B014

Store-set training raced a violation flush

Its own invariant on the first regression

B015

Slot-1-only load could miss the final LQ entry

Independent LQ model and one-free-slot reproducer

ACCELERATOR

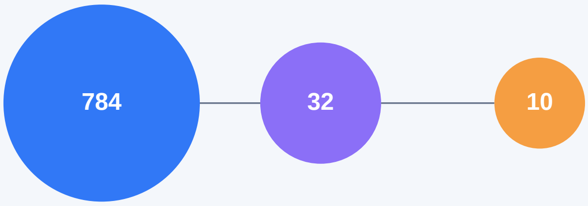
A real accelerator workload, not a toy MAC demo

The tightly coupled NPU executes the integer matrix work of a trained 784-to-32-to-10 MNIST MLP. CPU-enforced ordering makes accelerator access correct without software polling.

4x4 systolic array

int8 inputs and weights / int32 accumulators

Quantized MNIST MLP



97.13% integer accuracy / 10,000 offline images

Same workload: RV32I software vs NPU

Five-stage in-order

85.99x

96,367,418 software cycles -> 1,120,610 NPU cycles

Two-wide OoO

93.30x

58,535,712 software cycles -> 627,343 NPU cycles

32 / 32 bit-exact

Software and NPU logits are bit-exact on 32 of 32 exported images.

Board demo is built, not flashed

The switch-driven seven-segment board application self-tests 8 embedded images in simulation on both cores. Its 25 MHz image is assembled but has not run on silicon.

FPGA IMPLEMENTATION

Physical implementation: measured build, explicit silicon boundary

The current complete top includes the OoO core, NPU, initialized MNIST program and weights, PLL, timing constraints, and display MMIO. Quartus results and board results are shown as different evidence classes.

Current D029 routed top

34,945 / 49,760 LEs

15,140 registers / 632,444 memory bits

95 / 182 M9Ks / 16 multiplier elements

Actual PLL /2 place-and-route sign-off

25 MHz

Fmax 31.29 MHz

setup	**+8.045 ns**	hold	**+0.339 ns**
recovery	**+34.440 ns**	removal	**+1.506 ns**

Evidence classes

HARDWARE-CONFIRMED

In-order core

50 MHz

LED program running on the DE10-Lite; STA Fmax 53.95 MHz

HARDWARE-CONFIRMED

OoO core revision

16.67 MHz

M9K instruction fetch, LED walker, and standalone internal-flash boot on the board

BUILD-ONLY

D029 OoO + NPU + MNIST build

25 MHz

Placed, routed, timing-clean, assembled, unflashed

Implementation moves

- Banked M9K instruction memory with registered parity crossbar
- Gshare PHT moved from async logic to banked M9K storage
- PLL, complete SDC constraints, and all timing classes checked
- MIF-initialized data memory and MAX 10 ERAM configuration mode
- Six-read, three-write physical register file built as an M9K live-value table

ENGINEERING JUDGMENT

Decisions that show engineering judgment

The strongest part of the project is not feature count. It is the record of measured tradeoffs, rejected alternatives, invariants, and cycle-exact implementation changes.

D019

Balance the IQ select tree

PROBLEM

A 16-deep serial age scan limited the first OoO board port to 8.42 MHz.

CHOICE

Use a lower-index-preserving log-depth tree instead of registering the grant and adding scheduler latency.

EVIDENCE

Fmax 8.42 to 19.65 MHz, a 2.33x gain, with bit-identical IPC.

D021

Predict memory dependence with store sets

PROBLEM

Always-speculative loads improved CoreMark but made spill-heavy hello.c 36% slower.

CHOICE

Use SSIT plus LFST masks and train on real LQ violations; compare against conservative, always-speculate, and Alpha-style policies.

EVIDENCE

Recovered 90% of the spill regression and beat the Alpha-style policy by 12.7% on the precision microbenchmark.

D024-D025

Design storage for FPGA primitives

PROBLEM

Async PHT and 6R/3W PRF muxes consumed routing and logic needed by the full MNIST top.

CHOICE

Bank the PHT and implement the PRF as replicated M9K write banks with a live-value table.

EVIDENCE

The PRF change alone removed 8,895 LEs; the full board top moved from 88% to 70% logic use with cycle-identical behavior.

D029

Delete a bypass only after proving it dead

PROBLEM

The dmem-to-load-to-JALR bypass family became the routed critical path.

CHOICE

Prove the WB2 arm unreachable for every valid consumer, then keep the old mux as a permanent simulation oracle.

EVIDENCE

41,073,750 load writebacks and 47,050,722 tagged select targets in CoreMark produced zero WB2 uses; the 25 MHz build closes with +8.045 ns setup slack.

What I can defend in an interview

- Why IPC without Fmax is not performance on an FPGA
- Why synchronous block RAM changes pipeline contracts
- How to separate simulation, implementation, and silicon evidence without overclaiming
- How register renaming, recovery, SQ/LQ ordering, and store sets interact
- How lockstep, assertions, independent models, random stress, and coverag catch different bug classes

Reproduce it, inspect it, challenge it

The portfolio is a map to the evidence, not a replacement for it. The RTL evidence checkpoint and its passing CI run anchor the measured hardware state.

MEASURED SIMULATION

PROVES

Functional behavior, cycles, IPC, CoreMark CRCs, NPU equivalence, lockstep

DOES NOT PROVE

FPGA timing or physical operation

QUARTUS IMPLEMENTATION

PROVES

Fit, resource use, STA, configured clock margin, assembled image

DOES NOT PROVE

That the image has executed on the board

HARDWARE-CONFIRMED

PROVES

Observed behavior on the DE10-Lite at the stated clock and image

DOES NOT PROVE

A newer unflashed image or unsupported software stack

Evidence links

Repository

github.com/hannaashkar/rv32i_cpu

RTL evidence checkpoint

3c3171f

Passing production CI run

actions/runs/29368469898

Measured project brief

docs/PROJECT_BRIEF.md

Verification methodology

docs/VERIFICATION.md

Architecture decisions

docs/DECISIONS.md

Bug log

docs/BUGLOG.md

D029 proof and timing report

docs/WB_BYPASS_TIMING.md

Reproduction entry points

`make verify`
Full in-order system verification

`make verify-ooo`
OoO units, random stress, and recovery

`make coverage`
Combined both-core RTL line coverage

`make coremark-ooo`
Reportable 720-iteration OoO CoreMark run

`make npu-mlp-ooo`
Bit-exact software versus NPU MNIST run

`make synth-fit`
Quartus map and fit with logic-budget gate

Evidence boundary

25 MHz is build-proven, not yet silicon-proven. Linux is outside the implemented ISA and platform scope.

Hanna Ashkar

Electrical Engineering - Technion - computer architecture and FPGA systems

https://github.com/hannaashkar/rv32i_cpu

RTL evidence checkpoint 3c3171f

Architecture is only impressive when the measurements, failures, and proof survive scrutiny.